Anuj Pal

Decorah, IA | anujpald07@gmail.com | linkedin.com/in/anujpal | github.com/palan01-droid

SUMMARY

Computer Science sophomore and two-time IIT research intern, building full-stack and embedded systems from real-time IoT dashboards to multi-sensor biomechanics instrumentation. NCAA D-III varsity athlete with a track record of delivering under load. Seeking a **Summer 2027 internship**.

EDUCATION

Luther College — Aug. 2025 – May 2028 (Expected)
Bachelor of Arts in Computer Science — Decorah, IA

- **Relevant Coursework**: Data Structures, Algorithms, Python Programming, Discrete Mathematics

EXPERIENCE

Centre for Biomedical Engineering (CBME), IIT Delhi — Jun. 2026 – Present
Research Intern — Biomedical Instrumentation — New Delhi, India

- Built a real-time impact-testing system (**Python, Flask, Socket.IO**) that streams live force readings over WebSockets and **automatically captures peak pressure** on each impact for analysis
- Designed and wired a **multi-layer sensor circuit** with three FSR 402 sensors modeling skin, skull, and brain layers, measuring how impact force transfers through a physical head model
- Ramped from sensor and circuit fundamentals to a working data-acquisition pipeline within weeks, operating in a structured corporate research environment with regular progress reviews

Extreme Mechanics Laboratory, IIT Roorkee — Jun. 2026 – Jul. 2026
Research Intern — IoT & AI Systems — Roorkee, India

- Engineered an **IoT crash-detection device** for automotive safety under Prof. Shailesh Ganpule, building Arduino sensor circuits and Python pipelines for real-time impact-data collection
- Applied **machine learning** to crash-event data and built a live visualization GUI integrated with a companion mobile application

PROJECTS

Sensor Monitor | *Python (Flask, Socket.IO), Groq LLM API, Supabase, Docker/Render* — Live Demo · GitHub

- Full-stack platform streaming live sensor data from **4 microcontroller families** (Arduino, ESP32, STM32, Raspberry Pi Pico) to a 3D-visualized dashboard over WebSockets ($<$50ms latency); integrated the **Groq LLM API (Llama 3.3 70B)** to power a conversational AI that answers natural-language questions about sensor data and auto-explains anomalies
- Built a REST/WebSocket API with **Google OAuth via Supabase** supporting serial and WiFi sensor ingestion backed by PostgreSQL; **containerized with Docker and deployed live on Render**, backed by 23 pytest tests covering auth, ingestion, and validation

Drop Tower | *Python (Flask, Socket.IO), JavaScript (Chart.js), ESP32* — github.com/palan01-droid/drop_tower

- Open-source biomechanics dashboard streaming live FSR 402 readings from a 3-layer head model and **auto-logging peak impact values**; rolling charts visualize force propagation across skin, skull, and brain

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, HTML/CSS
Frameworks & Libraries: Flask, Socket.IO, Three.js, Chart.js, React.js, Node.js
Hardware & Embedded: Arduino, ESP32, STM32, Raspberry Pi Pico, Circuit Design, FSR Sensors
Data & AI: Machine Learning, Data Analysis, Real-Time Data Pipelines
Tools & Platforms: Git, GitHub, Linux/Bash, REST APIs, VS Code, AWS/GCP

ACTIVITIES & AWARDS

Men's Varsity Tennis, Luther College — American Rivers Conference Champions (2024–25); balanced NCAA D-III athletics with full-time CS coursework
Bronze Medal, SGFI National School Games — Lead Chandigarh nationals tennis team U-19; Ranchi, India (2024–25)